Name: _____ Register No: _____

Experiment 19

Question: Experiment to understand Logistic Regression for a given data set.

Aim

To understand the Logistic Regression algorithm and classify the Breast Cancer data set.

Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

def exp_19():
    from sklearn.datasets import load_breast_cancer
    from sklearn.model_selection import train_test_split
    from sklearn.preprocessing import StandardScaler
    from sklearn.linear_model import LogisticRegression
    from sklearn.metrics import (accuracy_score, confusion_matrix, classification_report,
                                 roc_curve, roc_auc_score)

    data = load_breast_cancer()
    X, y = data.data, data.target
    print("Dataset: Breast Cancer | samples =", X.shape[0], "| features =", X.shape[1])
    print("Classes: 0 =", data.target_names[0], ", 1 =", data.target_names[1])
    print("Class counts:", {str(n): int(c) for n, c in zip(data.target_names, np.bincount(y))},
        "\n")

    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.25, random_state=42, stratify=y)
    scaler = StandardScaler()
    X_train = scaler.fit_transform(X_train)
    X_test = scaler.transform(X_test)

    model = LogisticRegression(max_iter=1000)
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    y_prob = model.predict_proba(X_test)[:, 1]

    print("Train accuracy:", round(model.score(X_train, y_train), 4))
    print("Test accuracy :", round(accuracy_score(y_test, y_pred), 4))
    print("ROC-AUC score :", round(roc_auc_score(y_test, y_prob), 4))
    cm = confusion_matrix(y_test, y_pred)
    print("\nConfusion matrix:\n", cm)
    print("\nClassification report:\n",
          classification_report(y_test, y_pred, target_names=data.target_names))
    print("First 5 predicted probabilities of class 'benign':", y_prob[:5].round(3))
    print("First 5 predicted classes:", y_pred[:5], "| actual:", y_test[:5])

    fpr, tpr, _ = roc_curve(y_test, y_prob)
    fig, ax = plt.subplots(1, 2, figsize=(12, 4.8))
    sns.heatmap(cm, annot=True, fmt="d", cmap="Greens", xticklabels=data.target_names,
                yticklabels=data.target_names, ax=ax[0])
    ax[0].set(title="Confusion Matrix", xlabel="Predicted", ylabel="Actual")
    ax[1].plot(fpr, tpr, color="darkorange", label=f"ROC curve (AUC = {roc_auc_score(y_test,
        y_prob):.3f})")
    ax[1].plot([0, 1], [0, 1], "k--", label="Random guess")
    ax[1].set(title="ROC Curve", xlabel="False Positive Rate", ylabel="True Positive Rate")
    ax[1].legend()
    plt.tight_layout()
    plt.show()


exp_19()
```

Output

```
Dataset: Breast Cancer | samples = 569 | features = 30
Classes: 0 = malignant , 1 = benign
Class counts: {'malignant': 212, 'benign': 357}

Train accuracy: 0.9883
Test accuracy : 0.986
```

```
ROC-AUC score : 0.9977

Confusion matrix:
 [[52  1]
 [ 1 89]]

Classification report:
              precision    recall  f1-score   support

   malignant       0.98      0.98      0.98        53
      benign       0.99      0.99      0.99        90

    accuracy                           0.99       143
   macro avg       0.99      0.99      0.99       143
weighted avg       0.99      0.99      0.99       143

First 5 predicted probabilities of class 'benign': [0.969 0.    0.559 0.939 0.175]
First 5 predicted classes: [1 0 1 1 0] | actual: [1 0 1 1 0]
```

Graph

